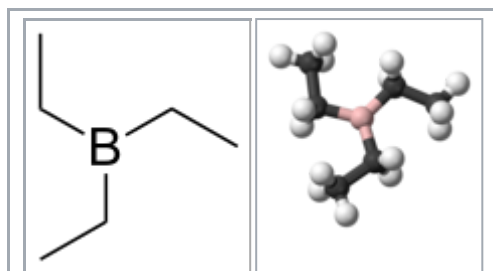




Triethylborane

Triethylborane



Names

Preferred IUPAC name

Triethylborane

Other names

Triethylborine, triethylboron

Identifiers

CAS Number

97-94-9 ([https://com
monchemistry.cas.o
rg/detail?cas_rn=97
-94-9](https://com
monchemistry.cas.o
rg/detail?cas_rn=97
-94-9)) ✓

3D model
(JSmol)

Interactive image ([https://chemapps.stol
af.edu/jmol/jmol.ph
p?model=B%28C
C%29%28CC%29C
C](https://chemapps.stol
af.edu/jmol/jmol.ph
p?model=B%28C
C%29%28CC%29C
C))

ChemSpider

7079 ([https://www.c
hemspider.com/Che
mical-Structure.707
9.html](https://www.c
hemspider.com/Che
mical-Structure.707
9.html)) ✓

ECHA
InfoCard

100.002.383 ([http
s://echa.europa.eu/
substance-informati
on/-/substanceinfo/1
00.002.383](http
s://echa.europa.eu/
substance-informati
on/-/substanceinfo/1
00.002.383))

EC Number

202-620-9

PubChem CID

7357 ([https://pubch
em.ncbi.nlm.nih.go
v/compound/7357](https://pubch
em.ncbi.nlm.nih.go
v/compound/7357))

UNII

Z3S980Z4P3 ([http
s://precision.fda.go
v](http
s://precision.fda.go
v))

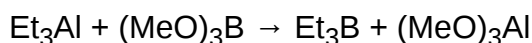
	v/uniisearch/srs/uni i/Z3S980Z4P3) ✓
CompTox Dashboard (EPA)	DTXSID2052653 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2052653)
InChI InChI=1S/C6H15B/c1-4-7(5-2)6-3/h4-6H2,1-3H3 ✓ Key: LALRXNPLTWZJIJ-UHFFFAOYS A-N ✓	
InChI=1/C6H15B/c1-4-7(5-2)6-3/h4-6H2,1-3H3 Key: LALRXNPLTWZJIJ-UHFFFAOYA U	
SMILES B(CC)(CC)CC	
Properties	
Chemical formula	(CH ₃ CH ₂) ₃ B
Molar mass	98.00 g/mol
Appearance	Colorless liquid
Density	0.677 g/cm ³
Melting point	−93 °C (−135 °F; 180 K)
Boiling point	95 °C (203 °F; 368 K)
Solubility in water	Not applicable; highly reactive
Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Spontaneously flammable in air; causes burns
GHS labelling:	
Pictograms	
Signal word	Danger

<u>Hazard statements</u>	H225, H250, H301, H314, H330, H360
<u>Precautionary statements</u>	P201, P202, P210, P222, P233, P240, P241, P242, P243, P260, P264, P270, P271, P280, P281, P284, P301+P310, P301+P330+P331, P302+P334, P303+P361+P353, P304+P340, P305+P351+P338, P308+P313, P310, P320, P321, P330, P363, P370+P378, P403+P233, P403+P235, P405, P422, P501
NFPA 704 (fire diamond)	
<u>Flash point</u>	< -20 °C (-4 °F; 253 K)
<u>Autoignition temperature</u>	-20 °C (-4 °F; 253 K)
<u>Safety data sheet (SDS)</u>	External SDS (https://www.sigmaaldrich.com/US/en/sds/aldrich/257192)
Related compounds	
Related compounds	Tetraethyllead Diborane Sodium tetraethylborate Trimethylborane
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✓ verify (what is ✓✗?) Infobox references	

Triethylborane (TEB), also called **triethylboron**, is an organoborane (a compound with a B–C bond). It is a colorless pyrophoric liquid. Its chemical formula is $(\text{CH}_3\text{CH}_2)_3\text{B}$ or $(\text{C}_2\text{H}_5)_3\text{B}$, abbreviated Et₃B. It is soluble in organic solvents tetrahydrofuran and hexane.

Preparation and structure

Triethylborane is prepared by the reaction of trimethyl borate with triethylaluminium.^[1]



The molecule is monomeric, unlike H_3B and Et_3Al , which tend to dimerize. It has a planar BC_3 core.^[1]

Applications

Turbojet engines

Triethylborane was used to ignite the JP-7 fuel in the Pratt & Whitney J58 turbojet/ramjet engines powering the Lockheed SR-71 Blackbird^[2] and its predecessor, the A-12 OXCART. Triethylborane is suitable because it ignites readily upon exposure to oxygen. It was chosen as an ignition method for reliability reasons, and in the case of the Blackbird, because JP-7 fuel has very low volatility and is difficult to ignite. Conventional ignition plugs posed a high risk of malfunction. Triethylborane was used to start each engine and to ignite the afterburners.^[3]

Rocketry

Mixed with 10–15% triethylaluminium, it was used before lift-off to ignite the F-1 engines on the Saturn V rocket.^[4]

The Merlin engines that power the SpaceX Falcon 9 rocket use a triethylaluminium-triethylborane mixture (TEA-TEB) as a first- and second-stage ignitor.^[5]

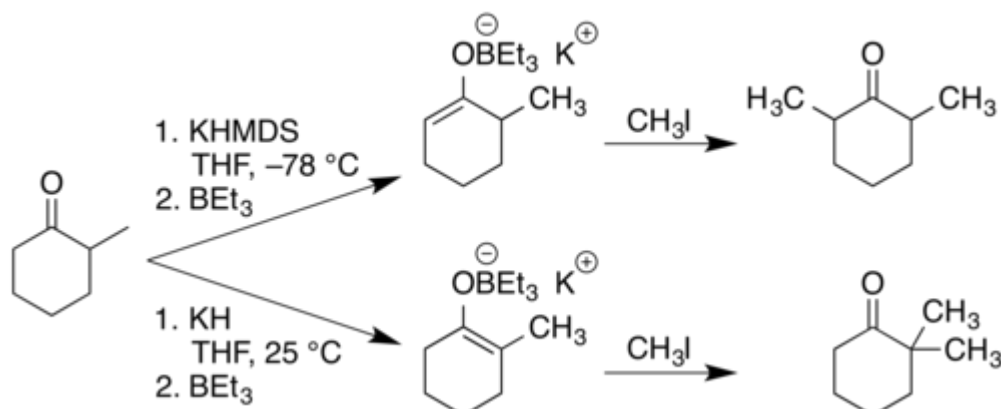
The Firefly Aerospace Alpha launch vehicle's Reaver engines are also ignited by a triethylaluminium-triethylborane mixture.^[6]

Organic chemistry

Industrially, triethylborane is used as an initiator in radical reactions, where it is effective even at low temperatures.^[1] As an initiator, it can replace some organotin compounds.

It reacts with metal enolates, yielding enoxytriethylborates that can be alkylated at the α -carbon atom of the ketone more selectively than in its absence. For example, the enolate from treating cyclohexanone with potassium hydride produces 2-allylcyclohexanone in 90% yield when triethylborane is present. Without it, the product mixture contains 43% of the mono-allylated product, 31% di-allylated cyclohexanones, and 28% unreacted starting material.^[7] The choice of base and temperature influences whether the more or less stable enolate is produced, allowing control over the position of substituents.

Starting from 2-methylcyclohexanone, reacting with potassium hydride and triethylborane in THF at room temperature leads to the more substituted (and more stable) enolate, whilst reaction at $-78\text{ }^{\circ}\text{C}$ with potassium hexamethyldisilazide, $\text{KN}[\text{Si}(\text{CH}_3)_3]_2$ and triethylborane generates the less substituted (and less stable) enolate. After reaction with methyl iodide the former mixture gives 2,2-dimethylcyclohexanone in 90% yield while the latter produces 2,6-dimethylcyclohexanone in 93% yield.^{[7][8]} The Et stands for ethyl group CH_3CH_2- .



It is used in the Barton–McCombie deoxygenation reaction for deoxygenation of alcohols. In combination with lithium tri-*tert*-butoxyaluminum hydride it cleaves ethers. For example, THF is converted, after hydrolysis, to 1-butanol. It also promotes certain variants of the Reformatskii reaction.^[9]

Triethylborane is the precursor to the reducing agents lithium triethylborohydride ("Superhydride") and sodium triethylborohydride.^[10]



Triethylborane reacts with methanol to form diethyl(methoxy)borane, which is used as the chelating agent in the Narasaka–Prasad reduction for the stereoselective generation of *syn*-1,3-diols from β -hydroxyketones.^{[11][12]}

Safety

Triethylborane is strongly pyrophoric, with an autoignition temperature of $-20\text{ }^{\circ}\text{C}$ ($-4\text{ }^{\circ}\text{F}$),^[13] burning with an apple-green flame characteristic for boron compounds. Thus, it is typically handled and stored using air-free techniques. Triethylborane is also acutely toxic if swallowed, with an LD_{50} of 235 mg/kg in rat test subjects.^[14]

See also

- Organoboranes
- Pentaborane
- Zip fuel

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